

INTRODUCTION TO BIOCHEMICAL ENGINEERING

Enzymes: Kinetics and Behavior in Biochemical Systems

Fundamentals of enzyme kinetics for bioprocess design and analysis

Introduction to Enzymes

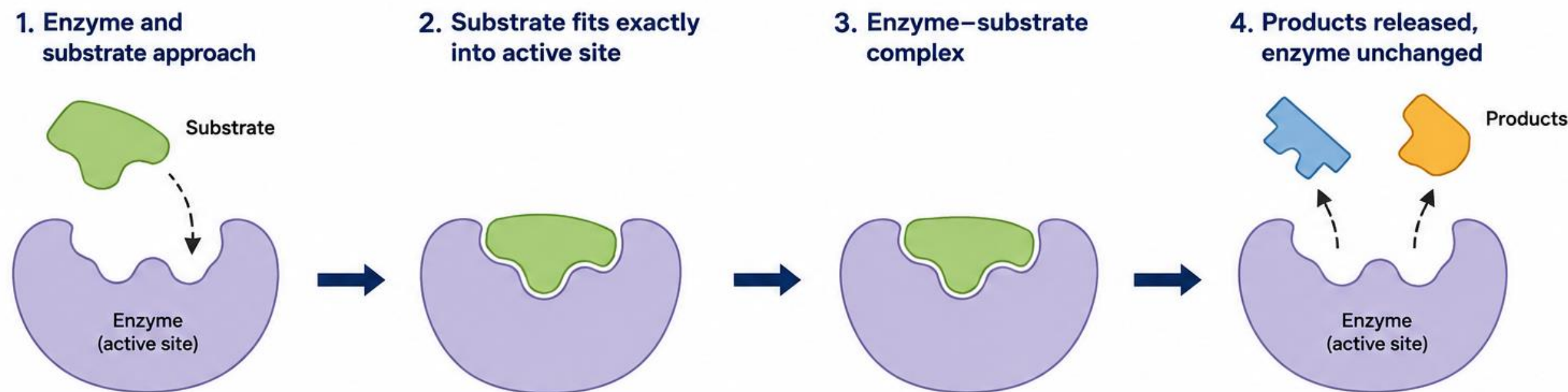
Enzymes are proteins that catalyze specific biochemical reactions without being consumed in the process.

KEY PROPERTIES

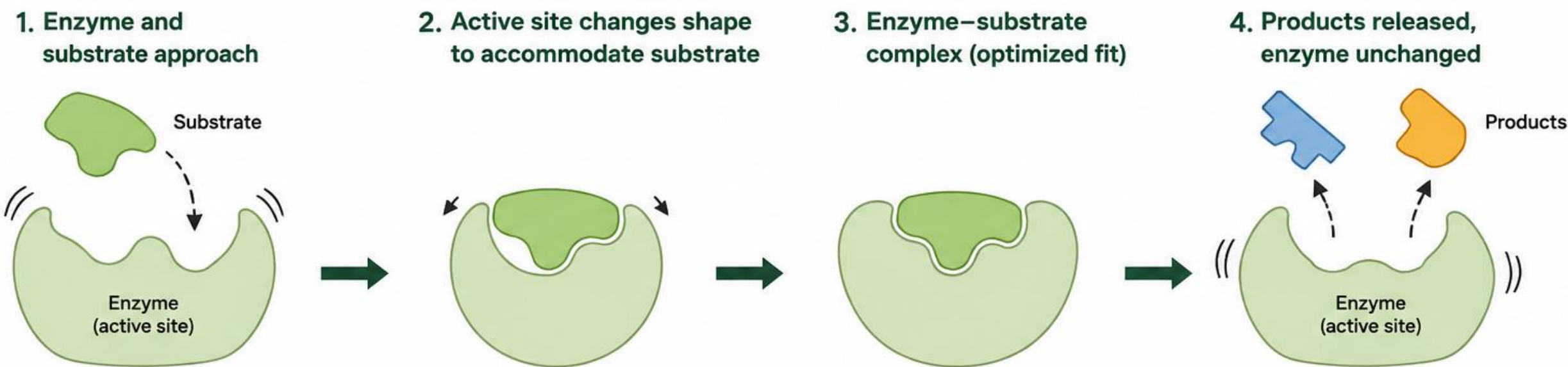
- Enzymatic reaction occur in specialized pockets called active sites.
- **Catalytic efficiency** — rate enhancements of 10^6 – 10^{17} over the uncatalyzed reaction
- **Mild operating conditions** — near-neutral pH, physiological temperature, aqueous media
- **Specificity** — recognize one substrate or a narrow class
- Enzymes are central to bioprocessing: fermentation, biocatalysis, diagnostics, and pharmaceutical synthesis.

Introduction to Enzymes

1. Lock-and-Key Hypothesis



2. Induced Fit Hypothesis



Types of Enzymes

Enzymes are classified by the type of reaction they catalyze

Class	Reaction Catalyzed	Industrial Example
Oxidoreductases	Electron transfer (oxidation–reduction)	Glucose oxidase — biosensors
Transferases	Transfer of a functional group between molecules	Transaminases — chiral amine synthesis
Hydrolases	Bond cleavage via hydrolysis	Lipases — detergent formulation
Lyases	Bond cleavage without hydrolysis or oxidation	Pectinases — juice clarification
Isomerases	Intramolecular rearrangement (isomerization)	Glucose isomerase — high-fructose corn syrup
Ligases	Bond formation coupled to ATP hydrolysis	DNA ligase — molecular biology / biotech

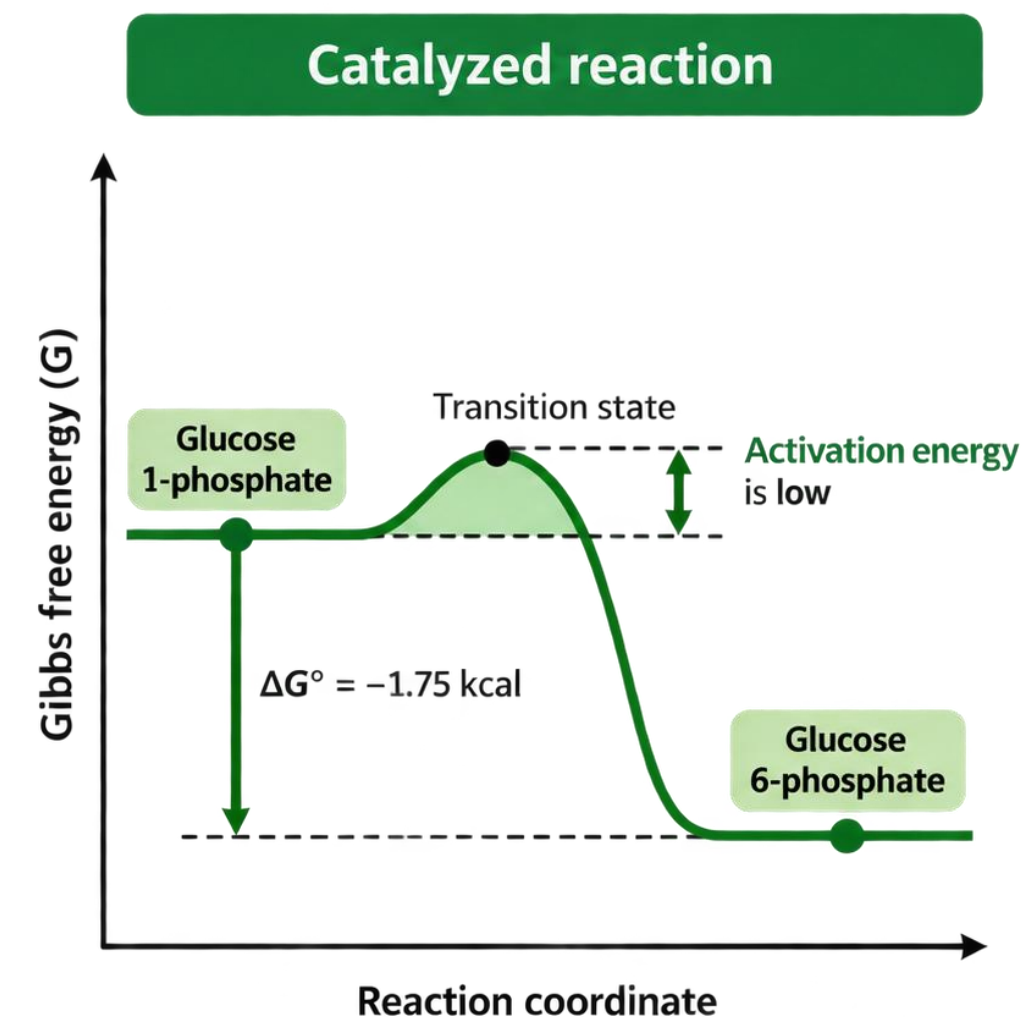
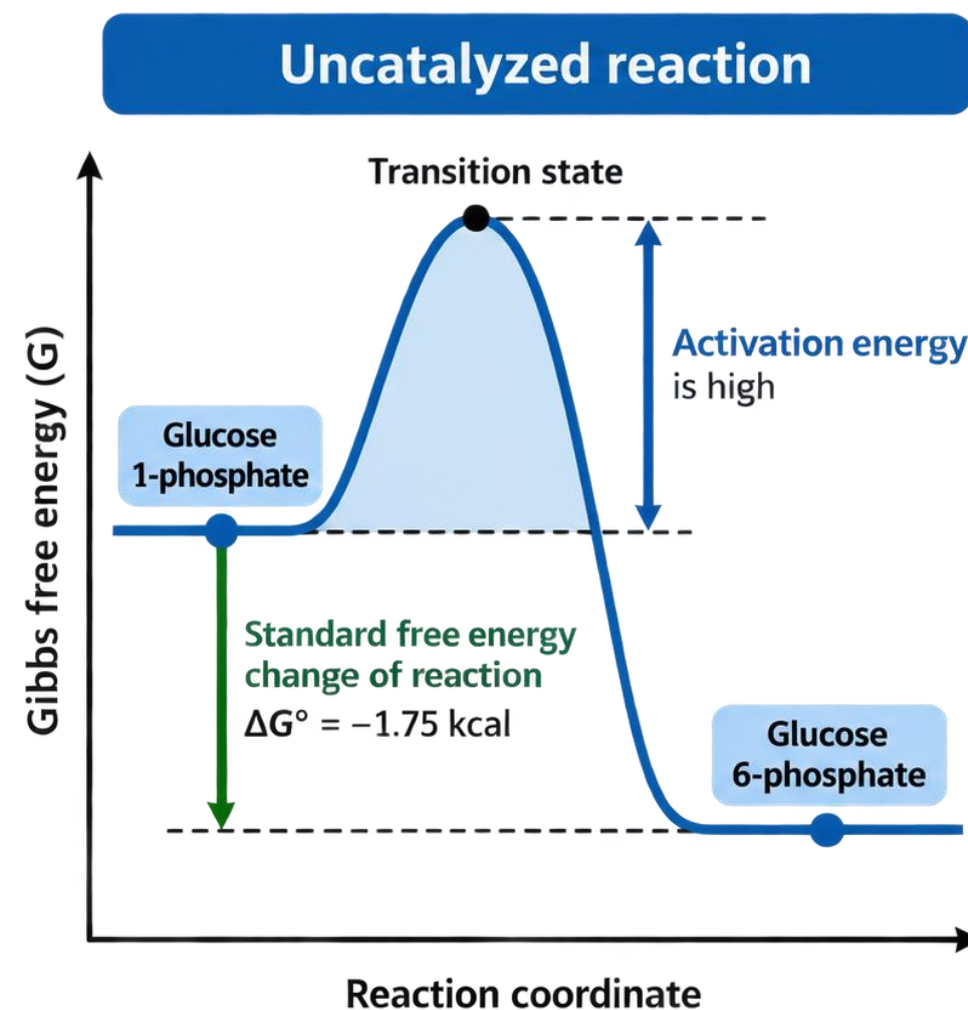
The Enzyme–Substrate Complex



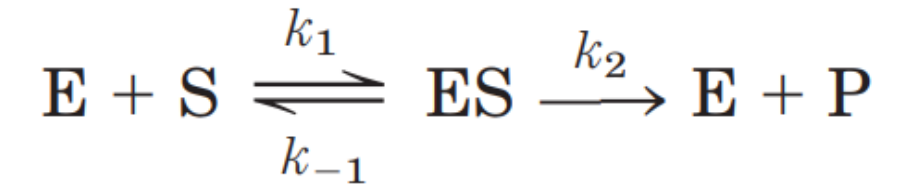
- Binding step k_1 and k_{-1} is reversible and typically fast relative to catalysis
- The ES complex juxtaposes substrate with catalytic residues, positioning it for reaction
- Breakdown of ES to product k_2 is commonly the slower, rate-limiting step — the basis for the steady-state treatment used to derive the rate law

Activation Energy

- Enzymes lower the activation energy E_a by stabilizing the transition state
- the overall free-energy change ΔG° is unchanged
- Rate follows the Arrhenius relation: $k = Ae^{-E_a/RT}$
- A small drop in E_a produces a large increase in k
- Stabilization arises from proximity and orientation effects, electrostatic complementarity, and substrate strain



Michaelis–Menten Equation



1. Determine reaction rate: $v = k_2[\text{ES}]$

2. Mass balance between ES: $\frac{d[\text{ES}]}{dt} = k_1[\text{E}][\text{S}] - (k_{-1} + k_2)[\text{ES}]$

3. Steady state assumption

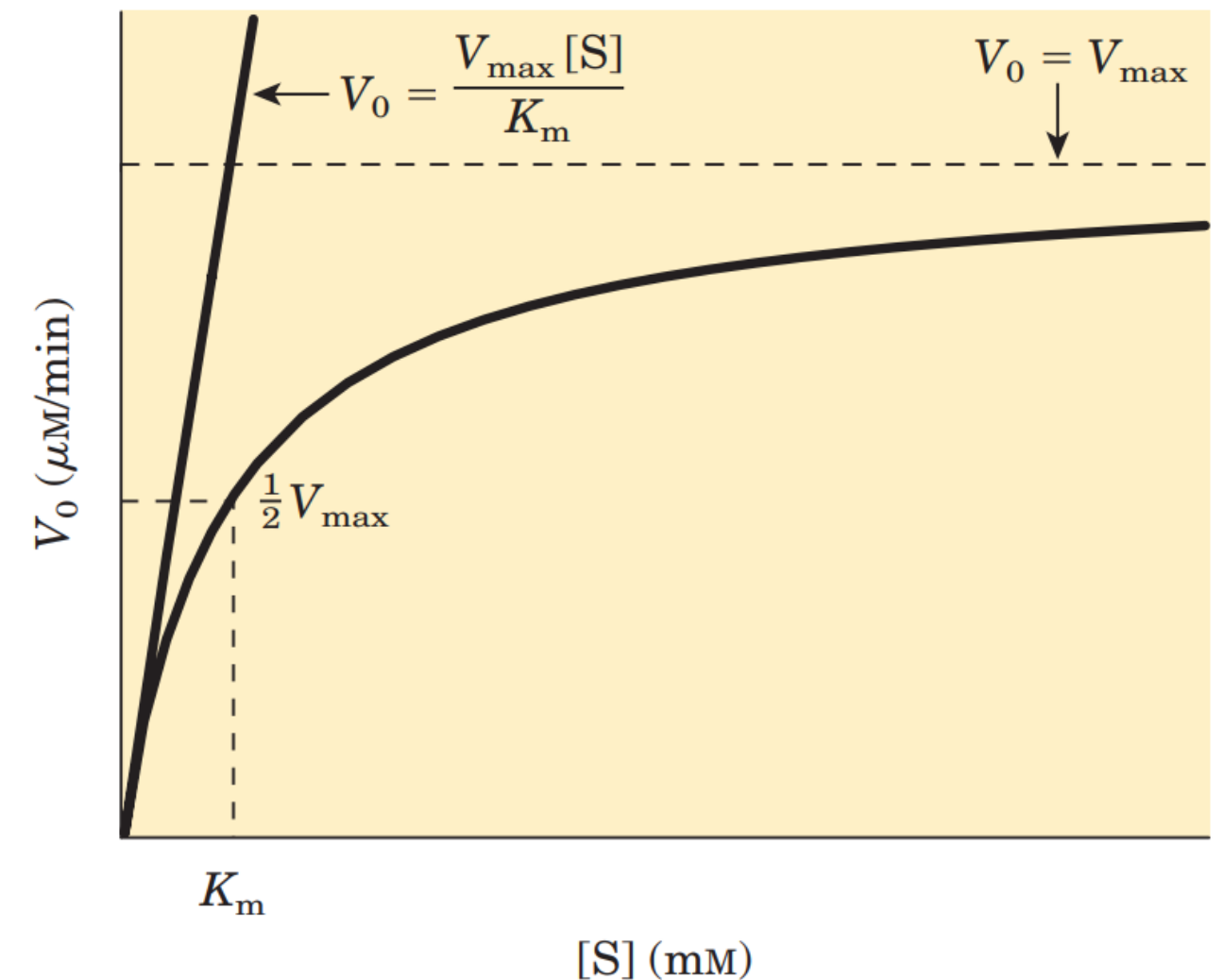
4. Michaelis-Menten constant:

$$K_m = \frac{k_{-1} + k_2}{k_1}$$

5. Enzyme mass balance: $[\text{E}]_T = [\text{E}] + [\text{ES}]$

6. Final equation:

$$v = \frac{V_{\max}[\text{S}]}{K_m + [\text{S}]}$$



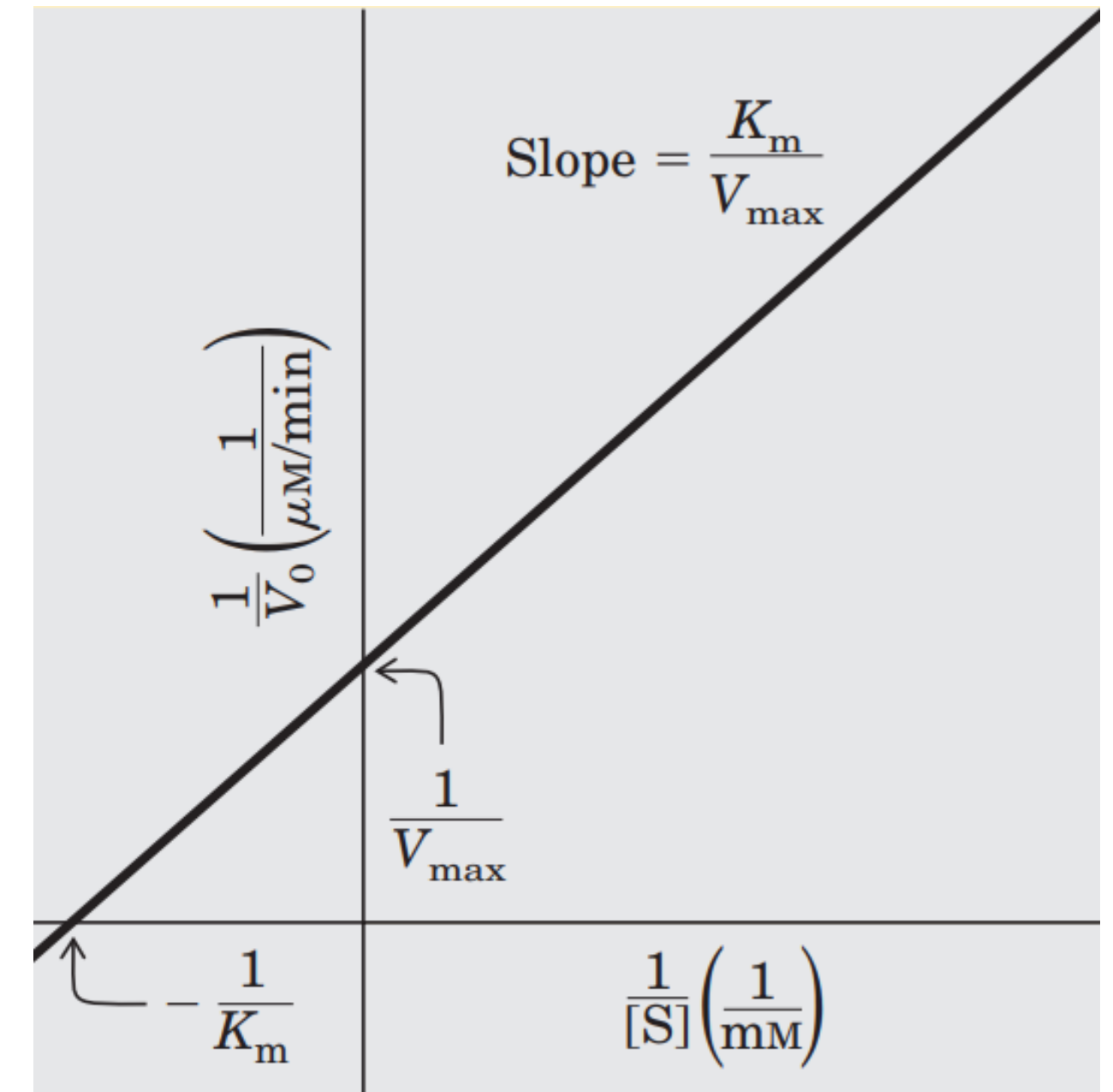
Lineweaver–Burk Plot

- Take the reciprocal of the Michaelis–Menten equation

- Final formula:

$$\frac{1}{v} = \frac{K_m}{V_{max}} \left(\frac{1}{[S]} \right) + \frac{1}{V_{max}}$$

- Weakness:** The reciprocal transform amplifies experimental error at low [S], where 1/v is large — those points disproportionately bias the fitted line.

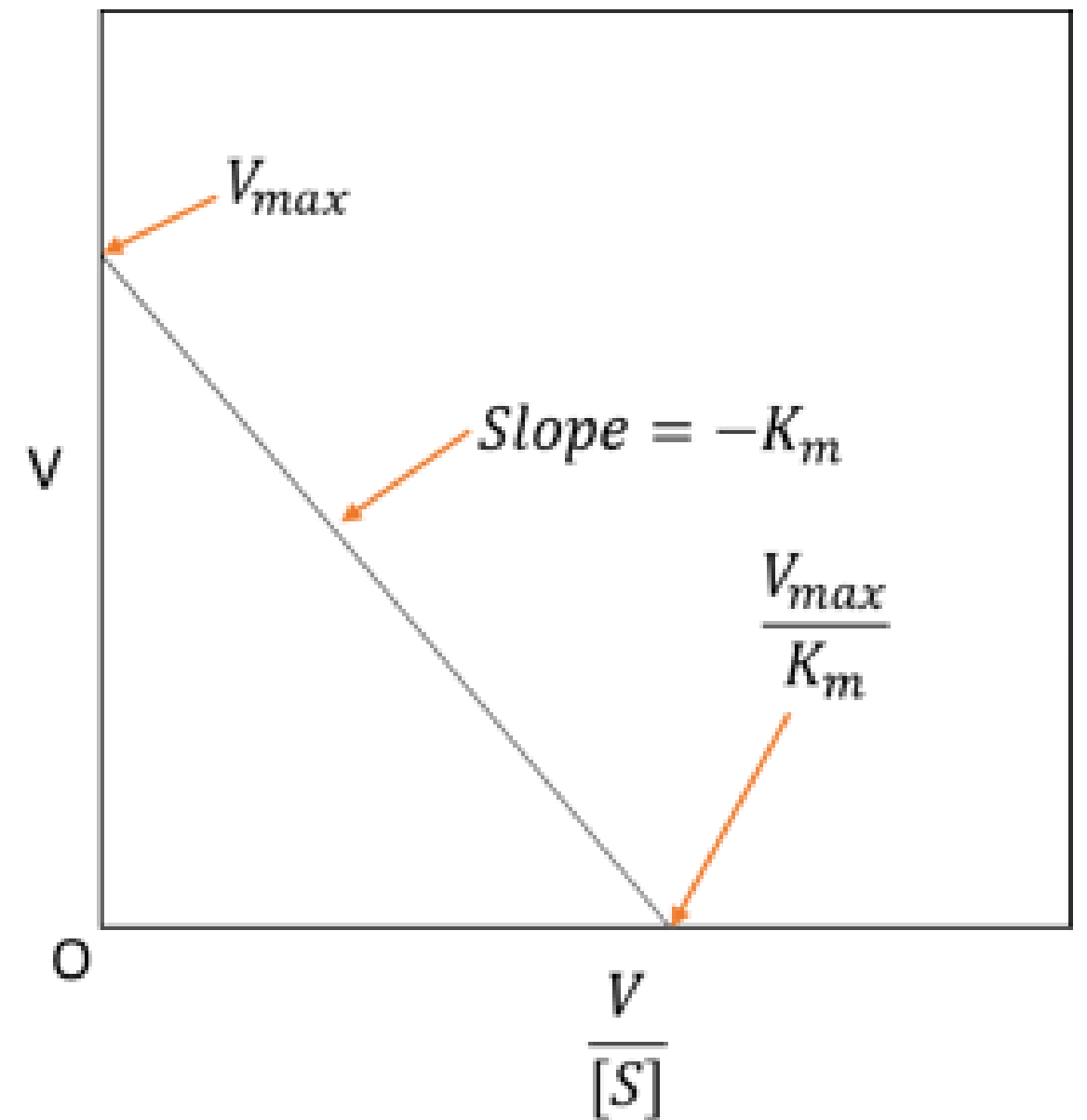


Eadie–Hofstee Plot

- Derived from the Michaelis-Menten equation to linearize enzyme kinetics data.
- It is written in the slope-intercept form -- $y = mx + c$

$$v = -K_m \left(\frac{v}{[S]} \right) + V_{\max}$$

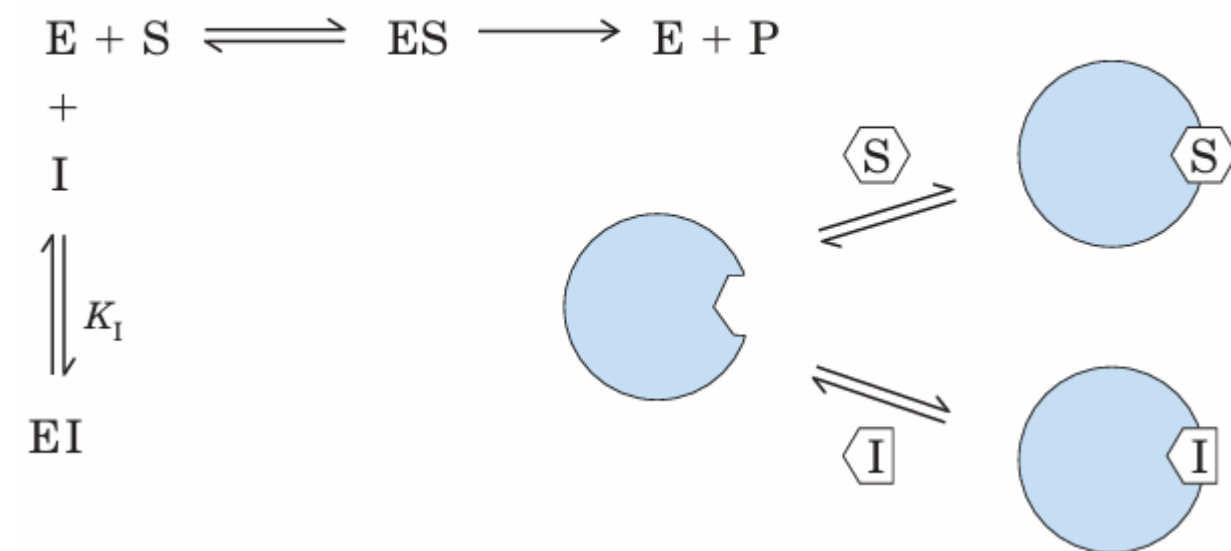
- Generally preferred over Lineweaver–Burk for parameter estimation, though v still appears on both axes and some correlation of error remains.



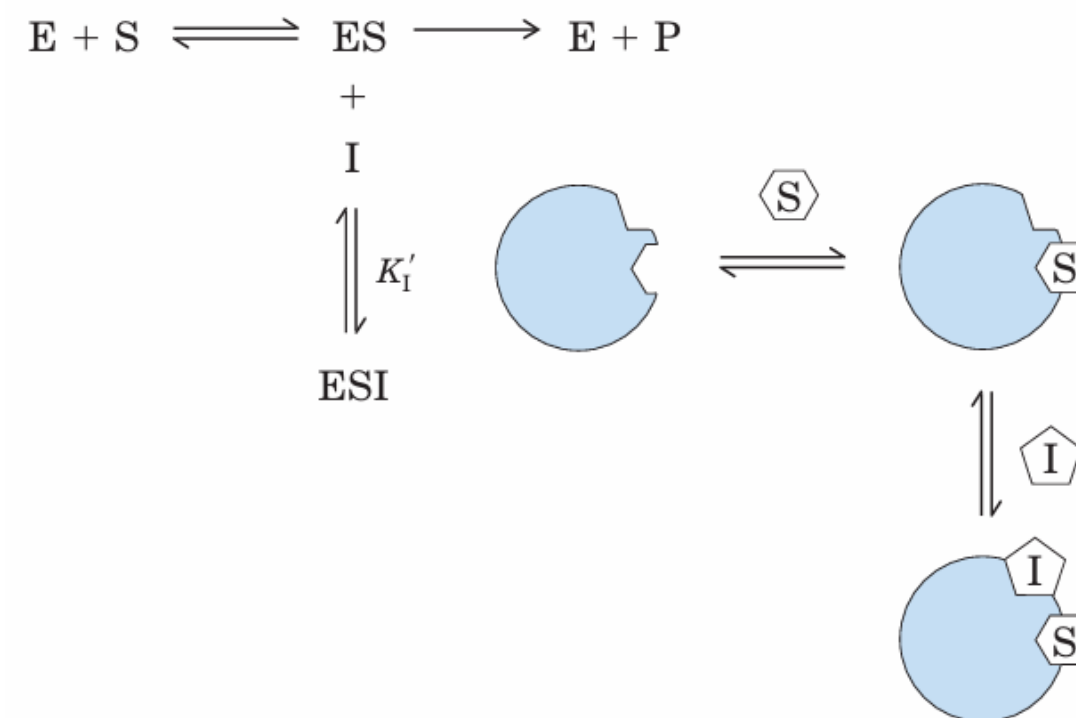
Product Inhibition

- Enzyme inhibitors: Molecular agents that interfere with catalysis – slowing or halting enzymatic reactions
- Two types:
 - Reversible Inhibition
 - Irreversible inhibition
- There are three types of reversible inhibitions: Competitive, Uncompetitive and Mixed competitive

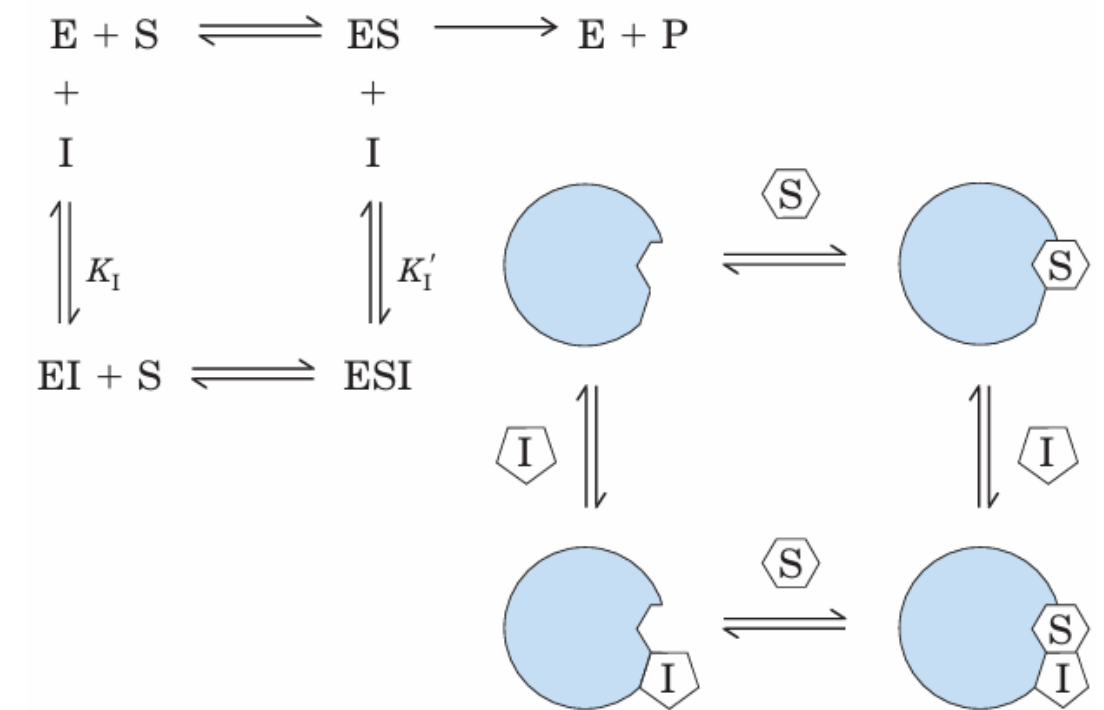
(a) Competitive inhibition



(b) Uncompetitive inhibition

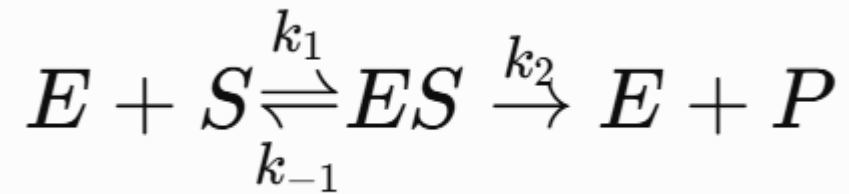


(c) Mixed inhibition

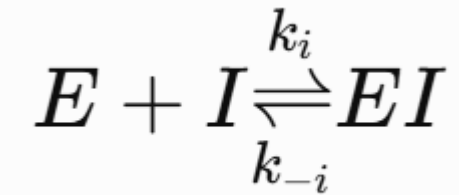


(d) Non-competitive Inhibition; I binds with E and ES equally.

Competitive Inhibition Enzyme Kinetics



and



1. Mass Balance on ES

$$\frac{d[ES]}{dt} = k_1[E][S] - (k_{-1} + k_2)[ES]$$

2. Mass Balance on EI

$$\frac{d[EI]}{dt} = k_i[E][I] - k_{-i}[EI]$$

3. Total enzyme mass balance

$$[E]_T = [E] + [ES] + [EI]$$

4. Finding E

$$[E] = \frac{[E]_T}{1 + \frac{[S]}{K_m} + \frac{[I]}{K_i}}$$

5. Find ES

$$[ES] = \frac{[E]_T[S]}{K_m \left(1 + \frac{[I]}{K_i}\right) + [S]}$$

6. Obtain Rate reaction

$$v = k_2[ES]$$

$$v = \frac{V_{\max}[S]}{K_m \left(1 + \frac{[I]}{K_i}\right) + [S]}$$

With $K_{m,\text{app}} = K_m \left(1 + \frac{[I]}{K_i}\right)$

Uncompetitive Inhibition Enzyme Kinetics



Objective: Find rate of reaction: $v = k_2[ES]$

1. Mass Balance on ES

$$\frac{d[ES]}{dt} = k_1[E][S] - k_{-1}[ES] - k_2[ES] - k_i[ES][I] + k_{-i}[ESI]$$

2. Mass balance on ESI

$$\frac{d[ESI]}{dt} = k_i[ES][I] - k_{-i}[ESI]$$

3. Enzyme mass balance

$$[E]_T = [E] + [ES] + [ESI]$$

and

$$\alpha' = 1 + \frac{[I]}{K_i}$$

4. Use Michaelis Menten equation

$$[ES] = \frac{[E]_T[S]}{K_m + \alpha'[S]}$$

5. Use rate reaction

6. Final formula for uncompetitive inhibition

$$v = \frac{V_{\max}[S]}{K_m + \alpha'[S]}$$

With

$$K_{m,\text{app}} = \frac{K_m}{\alpha'}$$

Type	Inhibitor binds to	K_m	$V_{\max}$	Can $\uparrow[S]$ overcome it?
Competitive	Free enzyme (E)	$\uparrow$	No change	Yes
Noncompetitive (pure)	E and ES equally	No change	$\downarrow$	No
Uncompetitive	ES complex only	$\downarrow$	$\downarrow$	No
Mixed	E and ES with different affinities	$\uparrow$ or $\downarrow$	$\downarrow$	No

Problems

1. Relation between Reaction Velocity and Substrate Concentration

- (a) At what substrate concentration would an enzyme with a k_2 of 30.0 s^{-1} and a K_m of 0.0050 M operate at one-quarter of its maximum rate?
- (b) Determine the fraction of $V_{\max}$ that would be obtained at the following $[S] = 1/2K_m$, $2K_m$, and $10K_m$

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1. Relation between Reaction Velocity and Substrate Concentration

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Solution:

a) According to the question: $v = 1/4 V_{\max}$ and $K_m = 0.0050 \text{ M}$

Substitute this in the Michaelis-Menten equation and simplify

Final answer: **$[S] = 1.67 \text{ mM}$**

b) Similarly substitute the different $[S]$

The fraction of $V_{\max}$ for each is **33.3%, 66.7%, 90.9%**

Problems

2. Although graphical methods are available for accurate determination of the $V_{\max}$ and K_m of an enzyme-catalyzed reaction, sometimes these quantities can be quickly estimated by inspecting values of V at increasing $[S]$.

Estimate the $V_{\max}$ and K_m of the enzyme-catalyzed reaction for which the following data were obtained.

$[S]$ (M)	V ($\mu\text{M}/\text{min}$)
2.5×10^{-6}	28
4.0×10^{-6}	40
1×10^{-5}	70
2×10^{-5}	95
4×10^{-5}	112
1×10^{-4}	128
2×10^{-3}	139
1×10^{-2}	140

Problems

Solution:

- As per the table, when [S] is increasing, V is not longer increasing
- This means that the enzyme is approaching saturation thus $V_{\max} \approx 140 \mu\text{M}/\text{min}$

$$v = \frac{V_{\max}}{2} \quad \text{when} \quad [S] = K_m$$

- Based on the above equation. We can calculate and find the corresponding [S].
- This [S] determines the K_m as well.

Problems

3. The half-maximal velocity of an enzyme-catalyzed reaction was found at a substrate concentration of 0.5×10^{-6} M. The enzyme follows Michaelis–Menten kinetics.

In the presence of a competitive inhibitor, the half-maximal velocity was found at 1.5×10^{-6} M. Given that the enzyme–inhibitor pair has a dissociation constant of 2×10^{-7} M, calculate the inhibitor concentration in μM .

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Solution:

- Without inhibitor, $[S] = 0.5 \times 10^{-6}$ M at half of V_{max}
- Thus $K_m = [S]$
- In presence of competitive inhibitor $[S] = 1.5 \times 10^{-6}$ M at half max velocity. Thus $K_{m,\text{app}} = [S]$
- For competitive inhibition:
$$K_{m,\text{app}} = K_m \left(1 + \frac{[I]}{K_i} \right)$$
- Substitute K_i and $K_{m,\text{app}}$ to find $[I]$
- Therefore concentration of
$$[I] = 4 \times 10^{-7} \text{ M}$$

Problems

4. An enzyme has $K_m=4 \text{ mM}$, In the presence of 2 mM of an uncompetitive inhibitor, the apparent K_m becomes $K_{m,app}=1 \text{ Mm}$. Calculate K_i' .

Problems

4. An enzyme has $K_m=4$ mM, In the presence of 2 mM of an uncompetitive inhibitor, the apparent K_m becomes $K_{m,app}=1$ mM Calculate K_i' .

Solution:

- For uncompetitive inhibition: $\alpha' = 1 + \frac{[I]}{K_i'}$

- Calculate α'

- Using:

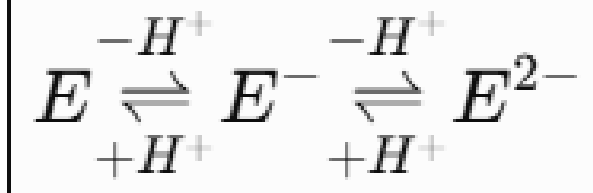
$$K_{m,app} = \frac{K_m}{\alpha'}$$

- Calculate K_i'

- Final:

$$K_i' = 0.67 \text{ mM}$$

Effect of pH



where:

$$K_1 = \frac{[E^-][H^+]}{[E]} \quad K_2 = \frac{[E^{2-}][H^+]}{[E^-]}$$

- Total enzyme concentration:

$$e_0 = e + e^- + e^{2-}$$

- Effect on Maximum reaction velocity:

- Fraction of active enzyme:

$$y^- = \frac{e^-}{e_0} = \frac{1}{1 + \frac{h^+}{K_1} + \frac{K_2}{h^+}}$$

$$v_{\max} = ke_0 y^-$$

- The active fraction reaches maximum:

$$h_{\text{optimum}}^+ = \sqrt{K_1 K_2}$$

- The optimum pH:

$$(pH)_{\text{optimum}} = \frac{1}{2}(pK_1 + pK_2)$$

$$v_{\max} = \frac{ke_0}{1 + \frac{h^+}{K_1} + \frac{K_2}{h^+}}$$